Develop a backup plan

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STEP 1

Identify the person, team, or device that will complete backups. For this activity, you have three choices:

- Store manager
    - IT team
- Automated backup system

Be sure to explain your choice.

Automated backup system

- Automated backups reduce the risk of human error, ensure backups are completed on schedule, and require minimal manual intervention. This guarantees reliability, standardization, and centralized logging. The IT team will monitor the backup system and resolve any issues that arise.

    - Relying on a Store Manager, it may introduce heavy risks of human error, forgetfulness, or local physical disasters.

STEP 2

Identify the types of data that the company should back up. Your answer must include one or more of the following types:

- Employee records
- Financial records
- Customer data
- Business documentation
- Intellectual property
- System data
- AI Models, Datasets & Vector Databases

List one example for each data type that you identify.

Employee Records

Example: Employee payroll records, Shift schedules, contracts, and performance reviews.

Financial Records

Example: Sales reports, Daily Point of Sale (POS) transaction logs invoices, tax records, and accounting data.

Customer Data

Example: Customer names, contact information, purchase history, and loyalty program records and encrypted payment tokens.

Business Documentation

Example: Company policies, supply contracts, operating procedures, and training manuals.

Intellectual Property

Example: Product designs, software source code, and marketing strategies.

System Data

Example: Server configurations, operating system files, databases, and application settings.

AI Models, Datasets & Vector Databases

Example: Fine-tuned model weights, training/validation datasets, prompt template repositories, and vector embeddings.

Step 3
Identify the type of daily backup that your company should perform:

- Full
- Differential
- Incremental

To make your choice, consider which backup type is best suited for a large retail franchise. Explain your choice.

The company should perform Incremental Backups every day.

Incremental backups save only the data that has changed since the previous backup. (such as today's new transactions and updated inventory counts) This method is ideal for a large retail franchise because it uses less storage space, completes quickly, and minimizes disruption to daily business operations. A full backup can then be performed weekly for complete system protection.

Step 4

Use the 3-2-1 rule to identify where your company should store its backups

and explain Storage Strategy for AI Systems.

Which types of storage will you use, and where will backups be located?

Be sure to justify your choices using the 3-2-1 rule.

The company will follow the 3-2-1 backup rule:

3 copies of all important data (one original and two backup copies).
2 different storage media, such as a local backup server and an external hard drive.
1 off-site copy stored securely in cloud storage.

This strategy protects data from hardware failure, accidental deletion, cyberattacks, theft, or natural disasters. If one backup is lost or damaged, the company can restore data from another copy.

For AI system:

Immutable Backups for AI Pipelines: Enforce **Write Once, Read Many (WORM)** cloud storage for training datasets and gold-standard model weights.

Reasoning: In the event of an AI **data poisoning attack** or unauthorized model alteration, immutable backups allow the security team to roll back the AI system to a clean, verified state.

Step 5

Determine when your company should perform backups: daily, weekly, or a combination of both.

Justify your choice.

The company should perform a combination of daily and weekly backups.

Daily incremental backups should occur automatically after business hours to capture new and updated data.

Weekly full backups should be performed every weekend to create a complete copy of all company data.

This schedule provides strong data protection while reducing backup time, minimizing storage requirements, and ensuring quick recovery in the event of data loss.